

Natnael Meseret

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SUMMARY

Full-Stack Software Engineer with over 6 years of experience building scalable distributed systems, real-time telecommunications infrastructure, and full-stack web applications. Specialized in Elixir, Go, TypeScript, React, and cloud integrations (AWS/GCP). Proven track record of architecting background processing pipelines, payment integrations, and complex API ecosystems.

WORK EXPERIENCE

Fullstack Software Engineer

Voiger

January 2025 - Present

- **FreeSWITCH ESL integration:** implemented a full call lifecycle event processing pipeline covering channel create, answer, bridge, hold/unhold, and hangup — with edge-case handling for transfers and multi-leg calls.
- **CDR tracking:** built multi-leg call detail record tracking with event-sourced talk time, hold time, queue time, and duration calculations, including linked child CDRs for transfers.
- **Campaign calling infrastructure:** designed and kickstarted the implementation of multi-modal outbound calling modes with SIP caller ID resolution, call recording, and per-campaign CDR creation.
- **Call routing:** built call routing features with various target type transfers, call recording, IVRs, timeout handling, configurable fallback targets, and call monitoring via eavesdrop and intercept and timeslot handling.
- **FreeSWITCH management:** built a FreeSWITCH management component using Horde dynamic supervisor to run API commands, manage inbound sessions across instances, and handle distributed process registration and failover across nodes.
- **Outbox sync system:** utilized Oban background jobs to reliably propagate agent, queue, and tier changes to FreeSWITCH in serial order with retry semantics.
- **Background workers:** built several Oban workers to run scheduled reports, free stuck resources, and perform periodic cleanup tasks.
- **Survey engine:** built per-call survey handling and completion tracking along with analytics APIs.
- **Keycloak integration:** integrated Keycloak for authentication and user session management.

EDUCATION

Addis Ababa University, Addis Ababa

September 2015 - June 2019

Bachelor of Science in
Computer Science

TECHNICAL SKILLS

Programming : Go, Elixir, JavaScript, TypeScript, Node.js, Lua

Frameworks/Libraries : Echo, React, Vue.js

Database : PostgreSQL, MySQL, MongoDB

Version Control : Git

Collaboration : Github, Gitlab

Operating Systems : Linux

Others : Firebase, Docker, RabbitMQ, GCP, AWS, Heroku

PERSONAL SKILLS

- Problem solving
- Organization
- Time management
- Communication
- Fast learner

LANGUAGES

- Amharic
- English

REFERENCE

Will be provided upon request.

- **RESTful APIs:** designed and implemented CRUD endpoints with moderately complex query filtering, multi-tenant scoping, input validation schemas, and database migrations.

Fullstack Software Engineer

ScrumLaunch LLC

Project : Nayak AI

February 2026 - June 2026

- **Stripe integration:** implemented subscription lifecycle flows, webhook handlers, customer creation and sync logic, and dynamic billing portal configurations.
- **Access control & restriction:** built automated subscription restriction behavior, read-only route guards for unpaid accounts, and grace-period account suspension flows.
- **AWS S3 integration:** setup S3 bucket configurations and implemented direct client uploads using presigned URLs with expiration.
- **Bulk data ingestion:** optimized bulk CSV ingestion pipelines using raw SQL prepared statements, paired with a frontend preview and inline editing interface prior to import.
- **Analytics & trend reporting:** implemented reporting modules aggregating weekly performance indicators to calculate and render trend directions.
- **LLM integration with LangChain:** setup automated meeting summarization pipelines, prompt optimizations to minimize token usage, and structured output extraction.
- **RESTful API design:** designed and implemented CRUD endpoints with query filtering, input validation schemas, and database migrations.
- **Background job processing:** utilized **BullMQ** to run background synchronization and lookup jobs asynchronously.
- **SendGrid integration:** setup dynamic SendGrid email templates and implemented backend payload delivery for automated notifications.

Fullstack Software Engineer

ScrumLaunch LLC

Project : Pittbos

July 2022 - December 2025

- Setup a **data migration flow** that ensures transition from a legacy structure to a desired structure for our relational database with options to rollback and make corrections if need be.
- Implemented Integration with **QuickBooks Online (QBO)** as an **OAuth 2.0 client** for syncing end users' customers, products & services, and invoice data with their QuickBooks accounts with version tracking across multiple qb accounts.

- Implemented **RESTful APIs** and wrote API and system **documentations**.
- Setup **role based access control (RBAC)** and APIs to manage roles and permissions from a management portal.
- Handled **deployment and monitoring** operations in AWS for **staging** environments and assisted in **production** environments.
- Implemented APIs with queries that run across **relational(postgres)** and **non-relational(mongodb)** databases to aggregate and fetch **system usage statistical data** displayed on a management portal.
- Implemented a utility function for **compressing and downsizing** user uploaded images when needed.
- Implemented a **file system watcher** service as an **OAuth 2.0 client** for automated transfer of ticket/receipt files from asphalt plants running **B&S Light'S EZ loader system**.
- Setup **GitLab CI pipeline** for **building and packaging** the service above as a **windows background startup service** for persistence and robustness.
- Implemented **APIs for exporting** formatted ticket/receipt data for **maestro*ERP's accounting tool**.
- Implemented **APIs for importing** bulky data from excel files with options to undo imports and make corrections if need be.
- Implemented product features across the stack, delivering backend functionality in Go and new UI components in React.
- Identified, fixed, and refactored issues in existing frontend codebases to improve stability, readability, and maintainability.
- Introduced background processing in the backend to handle heavy report aggregations and batch audit log loading from Redis-based queues to AWS DocumentDB, reducing AWS costs.

Senior Software Developer

Elnet Technologies Plc, Addis Ababa

July 2021 - November 2024 (Part-time)

- Implemented multiple payment method integrations for the **Taxiye** ride hailing app using **NodeJs**.
 - **HelloCash**
 - **M-PESA**
 - **PayPal**
 - **Amole** (through a payment aggregator platform)
 - **TeleBirr** (through a payment aggregator platform)
- Implemented backend for **TopUp**, **CashOut**, and **Payment** features using the above integrations for **Taxiye**.
- Implemented backend for payment method related settings.
- Implemented an automated lottery feature for **Taxiye**.

Senior Software Engineer

Derash Technologies, Addis Ababa

April 2020 - February 2022

- Built several UI components for the company website using VueJs and Bootstrap.
- Worked on a business process automation(BPA) SaaS platform called **Mengede** as a full stack developer. It automates the ticket sales process of intra-country bus companies and makes online reservation features available to customers.
- Built **RESTful API** backend for several resource management features.
- I implemented a real-time synchronization feature (between several ticket stations and online public users) using websockets. It synchronizes both seat interactions and reservations.
- Built a **mobile first** public portal for online seat reservations.
- I used GitLab as a repository hub and also configured it as a container registry for the project using the GitLab CI service.
- I used docker to containerize the project environment (Go application + MySQL database + Nginx reverse proxy for possible future sub services).
- I created a VM on AWS and configured it as a web server for the containerized project.
- I generated Letsencrypt SSL certificates for the VM using certbot.

Associate Software Engineer

Miltolabs(now Kolli Software. Eng & Cons. Plc), Addis Ababa

March 2020 - December 2020

- Built microservices that follow the **Command Query Responsibility Segregation(CQRS)** and **Event Sourcing** patterns.
- Utilized **MongoDB** as an eventstore for event sourcing pattern.
- Wrote service functions that publish and consume events to and from the eventbus(**RabbitMQ**).
- Used **Docker** to containerize the different microservices.

Software Engineer

Miltolabs(now Kolli Software. Eng & Cons. Plc), Addis Ababa

November 2019 - February 2020

- Built **Microservices** for a payment solution using Go.
- Created and managed **linux service files** for different microservices. This helps auto restart a microservice when it crashes.
- Installed and configured **Nginx** as a reverse proxy to act as an **API gateway**.
- Developed web services using **REST** architectural style.

- Built **connector modules** for two core banking systems(**Temenos T24** and **FlexCube**) using Go.
- Worked on **frontend portals** that interact with REST APIs using VueJs.
- Wrote JavaScript functions that utilize the **firebase** API for some of the features we had.
- Built a **websocket** backend that serves updates using Golang and frontend components that interact with it using VueJs.
- Created **diagrams** and **mind maps** that illustrate **system architectures** and interaction patterns.

Intern Junior Software Engineer

MiltoLabs(now Kolli Software. Eng & Cons. Plc), Addis Ababa

July 2019 - October 2019

- Got training on Overview of Software Engineering, **Go**, **Echo**, and **VueJs**.
- Introduced the team with a mind map tool called **Coggle** which is being used as a communication tool.
- Configured **firewall** and **DNS** on **google cloud platform**.
- Setup a VM on GCP as an initial deployment environment by installing the Go environment and configuring environment variables.
- Built the company website using VueJs.

Intern Software Developer

Sheger Mass Transport Service Enterprise, Addis Ababa

June 2018 - August 2018

- Developed a web based ticket management system.
- Worked as a full stack developer in the team.